

GG Calendar AI – Target System Plan (v2)

This document defines the refined system plan for GG Calendar AI. The system is designed as an intelligent web-based assistant that converts images, text, and chat instructions into draft calendar events, which are reviewed by the user and then committed to Google Calendar. Google Calendar remains the system of record and handles all notifications.

1. Core Philosophy

- 1 Google Calendar is the system of record.
- 2 The app never silently modifies calendars.
- 3 All changes flow through a draft-review-commit pipeline.
- 4 Users define their own schedules (unlimited, named freely).

2. Target User Experience

Users interact with GG Calendar AI through a web application with a chat interface. They can upload images of schedules, type natural-language commands, or ask questions about their existing calendar. The assistant generates drafts, shows them for confirmation, and only writes to Google Calendar after explicit approval.

3. User-Defined Schedules

Schedules are user-created containers (e.g., 'Classes', 'Work', 'Gym', 'Deadlines'). Each schedule maps to a dedicated Google Calendar. This separation enables clean organization and allows users to configure notification behavior directly within Google Calendar.

- 1 Unlimited schedules per user.
- 2 Each schedule stores a Google Calendar ID.
- 3 Users manage notification defaults inside Google Calendar.

4. Draft Generation Pipeline

- 1 Text input: parse intent, dates, times, recurrence, and target schedule.
- 2 Image input: OCR + layout interpretation to extract timetable structures.
- 3 Normalization: convert all inputs into structured event candidates.
- 4 Draft events stored locally and shown to user for review/editing.

5. Calendar Commit Service

The calendar commit service is responsible for writing approved draft events to Google Calendar. Before committing, the service performs conflict checks and deduplication. After successful creation, Google event IDs are stored for future reference.

- 1 FreeBusy checks prevent unwanted overlaps.
- 2 Import history prevents duplicate inserts.
- 3 Commit is explicit and user-triggered.

6. Notifications & Alerts Strategy

GG Calendar AI relies on Google Calendar's built-in notification system. Users configure reminders at the calendar or event level inside Google Calendar. Events created by the app inherit these settings automatically.

- 1 No custom notification infrastructure in v1.
- 2 Optional per-event reminders if explicitly requested.
- 3 Weekly schedule review can be implemented as a recurring Google event.

7. Schedule Queries & Chat Q&A;

The assistant can answer natural-language questions about the user's calendar, such as daily agendas, time-window queries, upcoming events, and availability. These queries are read-only and do not modify Google Calendar.

- 1 Examples: 'What do I have Friday?', 'From 5–9 what do I have?'
- 2 Queries can target one or multiple schedules.
- 3 Google Calendar data is always the source of truth.

8. Data & Persistence

The system uses a relational database to store users, schedules, OAuth tokens, drafts, draft events, and import history. SQLite is sufficient for private beta; Postgres is recommended for scaling.

9. Phased Roadmap

- 1 Phase 1: OAuth, schedules, draft → commit, event queries.
- 2 Phase 2: Conflict handling, deduplication, weekly reminders.
- 3 Phase 3: Image-based schedule imports.
- 4 Phase 4: Analytics and insights (optional).

10. Next Steps

With this plan validated, the next phase is to assess the developer's current setup and skills. Based on that assessment, a step-by-step implementation checklist (tools, installs, configuration) will be produced.